

Analysis of naïve lung CD4 T cells provides evidence of functional lung to lymph node migration

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The proportion of CD4 T cells with phenotypic and functional properties of naïve cells out of total CD4 T cells is similar in the lung parenchyma and lymph nodes. On treatment with a sphingosine-1-phosphate agonist, the frequency of these cells falls precipitously, but with a delay of ~14 h compared with blood CD4 T cells; neither anti-CD62L nor pertussis toxin prevents entry of naïve CD4 T cells into the lung. Based on treatment with anti-CD62L and the use of CCR7^{-/-} cells, lung naïve CD4 T cells appear to migrate to the mediastinal lymph nodes along a CD62L-independent, CCR7-dependent pathway. Cells that have entered the node in this manner are competent to respond to antigen. Thus, a portion (approximately one-half) of naïve CD4 T cells appears to enter the mediastinal lymph nodes through a blood-to-lung-to-lymph node route.

immune response | lymphocyte traffic | pulmonary lymphocyte

Naïve CD4 T lymphocytes are generally considered part of the recirculating pool and are thought to be largely confined to secondary lymphoid organs, trafficking among one another through lymph and blood (1, 2). After entering lymph nodes, these cells take an estimated 10–12 h to exit through efferent lymphatics and eventually enter the blood (2–4). This process is controlled by the expression of sphingosine-1-phosphate (S1P) receptors on the emigrating lymphocytes and the existence of a concentration gradient of S1P between lymph and lymph nodes (5).

Assuming a short interval from exit from the lymph node to reentry through high endothelial venules (for lymph nodes) or the splenic arterioles, an individual naïve CD4 T cell may make two or three complete circuits per day (6). Although the number of lymphocytes entering any given lymph node during steady state, before local inflammatory responses develop and lymphatic blood supply increases (7), is not precisely known, it is presumably a function of the steady-state blood supply to the node and related to node size. Thus, tissues drained by small lymph nodes will receive only a small fraction of the total naïve pool in the course of a single day, and consequently only a small proportion of the total naïve cells specific for any given antigen. Although it is known that the inflammatory response ensuing after the onset of immune responses can “open” the lymph node to other means of entry, and that blood supply is increased during immune responses (7, 8), conventional traffic patterns should be relatively inefficient in bringing antigen-specific naïve CD4 T cells to the site of initial priming at the outset of antigen challenge. This seems to be an issue particularly for mouse lung tissue, a frequent site of infection drained by the mediastinal lymph nodes (3, 4, 9).

Here we report that the lung contains a substantial number of naïve phenotype CD4 T cells. Indeed, the presence of naïve phenotype CD4 and CD8 in the lung and other tissues has been described previously (10). We provide evidence that these cells enter from the blood and traffic through the lung. They supply the draining lymph nodes through a CD62L-independent, CCR7-dependent pathway, presumably increasing the proportion of naïve CD4 “lymphons” that can be rapidly sampled by local antigen-presenting cells. We show that even when anti-CD62L is administered, mediastinal lymph node CD4 T cells can respond to intratracheal immunization, suggesting that this pathway is competent to supply antigen-responsive cells to the draining nodes.

Results

Phenotypic Analysis of Lung CD4 T Cells. A phenotypic analysis of CD4 T cells obtained from the lungs of normal 6- to 8-wk-old mice revealed the surprising result of naïve phenotype in the majority of cells. We used the degree of CD44 expression as an initial criterion to differentiate memory and effector phenotype CD4 T cells from naïve phenotype cells (11). Strikingly, the majority of lung CD4 T cells were CD44^{low}; the distribution of CD44^{low} and bright cells was indistinguishable from that in the mediastinal lymph nodes (Fig. 1A). We had exhaustively perfused the lung before analysis; virtually no red blood cells were detectable, implying that peripheral blood lymphocytes did not make a major contribution to the lung lymphocyte population under study.

L-selectin (CD62L) is expressed on naïve and central memory cells (7). Analysis of its expression in conjunction with CD44 allows estimation of the relative frequency of naïve (CD62L^{hi}/CD44^{low}), effector memory/effector (CD62L^{low}/CD44^{hi}), and central memory (CD62L^{hi}/CD44^{hi}) cells. No difference was observed in the distribution of these populations among lung and mediastinal lymph node CD4 T cells (Fig. 1A). In both cases, the main population had a naïve phenotype.

The distribution of expression levels of CD45RB, distinguishing naïve and memory phenotype cells (12), was also very similar in the lymph nodes and the lungs (Fig. 1A). IL-7R α (CD127) is expressed uniformly on naïve CD4 T cells in the periphery and is also required by memory cells for long-term survival (13, 14). IL-7R α expression is similar among lung and mediastinal lymph node CD4 T cells and is similar to that of peripheral lymph nodes in general.

We examined expression of the GPI-anchored membrane protein Ly6C (15) on naïve phenotype (CD44^{low}) and memory phenotype (CD44^{hi}) CD4 T cells. The naïve CD4 T-cell compartment in the lungs and lymph nodes contains relatively equal numbers of Ly6C^{hi} and dull cells. In contrast, the CD44^{hi} cells were uniformly Ly6C^{low} in both the lungs and the mediastinal nodes (16) (Fig. 1A).

Thus, using a set of markers that generally distinguish naïve phenotype and memory/effector phenotype cells and their subsets, we could find no difference between cells in the mediastinal lymph nodes and those in the lungs. In general, the mediastinal lymph nodes did not differ from other peripheral lymph nodes.

Recent Thymic Emigrants in the Lungs. These results suggest the existence of a dominant population of CD4 T cells in the lungs that is phenotypically indistinguishable from the naïve phenotype CD4 T cells of the peripheral lymph nodes. To explore this possibility further, we asked whether the proportion of recent thymic emigrants (RTEs) was similar in naïve phenotype cells of the lungs and mediastinal lymph nodes. To measure the frequency of RTEs in the lymph nodes and lungs, we used BAC-transgenic mice in which the gene encoding GFP had been recombined into the *Rag2* gene (*RAG2p-GFP*) (17). In such mice, GFP expression is maintained

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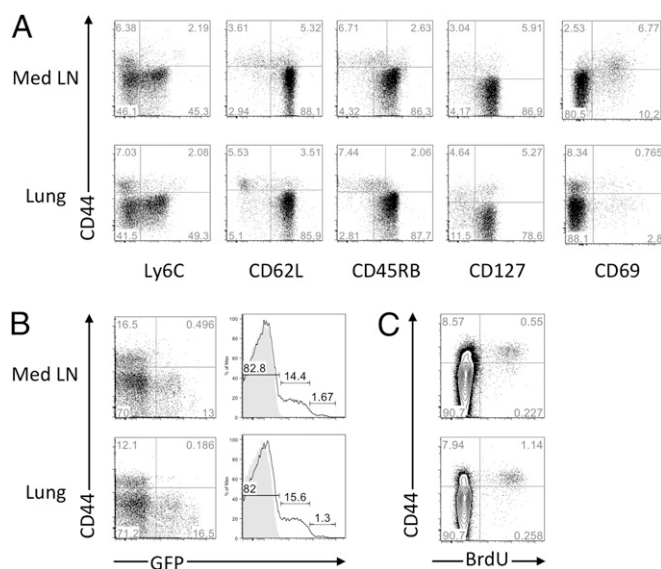


Fig. 1. (A) Phenotypic analysis of lung and mediastinal lymph node (Med LN) CD4 T cells from normal B10.A mice. Lung and lymph node lymphoid cells were gated on CD4⁺ and analyzed for surface expression of CD44, CD62L, CD45RB, IL-7R α (CD127), CD69, and Ly6C. (B) Frequency of recent thymic emigrants among lung and lymph node naïve phenotype CD4 T cells. Mediastinal lymph node and lung CD4 T cells from RAG2p-GFP and control mice were analyzed for expression of GFP on total CD4 T cells (Left) and on CD44^{low} CD4 T cells. Results from RAG2p-GFP are shown in solid outline; controls are in solid gray. (C) Normal B10.A mice were given 500 μ g of BrdU i.p. and analyzed 6 h later. Lungs and mediastinal lymph nodes were stained for CD4, CD44, and BrdU.

transiently in peripheral T cells, marking RTEs (18, 19). In the RAG2p-GFP mice, three different subsets of CD4 T cells within the naïve CD44^{low} subset can be identified: GFP^{hi} peripheral T cells that have left the thymus within the previous week, GFP^{low} cells that are 1–2 wk older, and GFP⁺ cells that are the fully mature peripheral naïve T cells (20). The frequency of GFP^{hi} and GFP^{low} peripheral T cells is reportedly similar in the spleen, blood, and lymph nodes. Approximately 1.5% of naïve phenotype CD4 T cells in the mediastinal lymph nodes and lungs are GFP^{hi} and ~15% are GFP^{low} (Fig. 1B), suggesting a similar frequency of RTEs among lung and lymph node CD4 T cells.

This conclusion is also supported by the transfer of CD4 single positive thymocytes from normal CD45.1 congenic B10.A mice into CD45.2 hosts (Fig. S1). At 3 d after transfer, CD45.1 CD4 cells were detected both in the lungs and in the mediastinal and popliteal lymph nodes. These cells retained a naïve phenotype (CD62L^{hi}, CD45RB^{hi} CD44^{low}). The finding that recent thymic emigrants are as frequent among naïve phenotype CD4 T cells in the lung as they are in the mediastinal lymph nodes strengthens the contention that the naïve phenotype cells of lung and lymph node are similar.

Naïve Phenotype Lung CD4 T Cells Do Not Proliferate Rapidly or Display an Activated Phenotype. Naïve phenotype CD4⁺ cells in the secondary lymphoid organs have a very low proliferative rate in contrast to memory phenotype cells in the mouse, which divide rapidly at steady state (21, 22). We previously showed that a 24-h “pulse” of BrdU results in ~10% of memory phenotype cells taking up the nucleotide, whereas ~0.1% of naïve phenotype cells become BrdU⁺ during a 24-h pulse (23, 24). Few if any naïve phenotype lung CD4⁺ T cells incorporate BrdU within 24 h, whereas 12% of the CD44^{hi} memory phenotype cells became BrdU⁺ (Fig. 1C). Similarly, the low proportion of CD69⁺ cells in lung CD44^{low} CD4 T cells is correlated with the characteristics of the naïve phenotype mediastinal lymph node CD4 T cells (Fig. 1A) (25–27).

Production of Effector Cytokines. Stimulation of lung cells ex vivo with phorbol 12-myristate 13-acetate and ionomycin followed by staining for intracellular IFN γ , IL-4, IL-13, and IL-17 revealed few if any positive cells among the lung or mediastinal lymph node naïve phenotype cells. In contrast, cells capable of secreting these cytokines were found among memory phenotype cells in the lungs (Fig. 2A and B). We examined the frequency of T-bet⁺ cells, as determined by ZS-Green expression among CD4 T cells from a BAC-transgenic reporter mouse (28). Whereas memory phenotype cells in the lungs contained a proportion of cells that were extremely bright for ZS-Green, virtually none of the naïve phenotype lung CD4 T cells or mediastinal lymph node CD4 T cells were ZS-Green⁺ (Fig. 2C). Interestingly, the mediastinal lymph node memory phenotype cells that were ZS-Green⁺ had a lower mean fluorescence intensity compared with lung memory phenotype cells, suggesting a larger proportion of effector memory or effector cells among the lung memory phenotype cells. We also examined expression of an IL-17 surrogate (IL-17F-RFP) and found that virtually none of the lung naïve phenotype CD4 T cells were RFP⁺ (29). Thus, based on functional properties as well as phenotypic markers, the majority of CD4 T cells in the lung can be classified as naïve.

Localization of Lung Naïve CD4 T Cells. To determine the location of naïve CD4 T cells in the lung, we transferred 2×10^6 carboxy-fluorescein succinimidyl ester (CFSE)-labeled CD4 lymph node cells from TCR transgenic 5C.C7 Rag2^{-/-} or GFP⁺ P25 Rag1^{-/-} donors (30, 31). The CD4 T cells in these mice are essentially all naïve and compose ~99% of the transferred cells. At 18 h later, the mice were killed, and frozen sections of perfused lungs were prepared. The naïve T cells were found in the same location as endogenous T and B cells in the lung (Fig. 3C and F) and were not seen within the CD31⁺ blood vessels (Fig. 3A, B, D, and E), implying the presence of cells in the lung parenchyma.

Naïve Phenotype Lung CD4 T Cells Are Depleted, with a Delay, by an S1P Agonist. Naïve unstimulated lymphocytes spend ~10–12 h within a lymphoid organ (32) before returning to the circulation. These cells recirculate through the lymph to the blood in the case of lymph nodes and Peyer’s patches, or directly to the blood from the spleen (33). We used treatment with the S1P receptor agonist

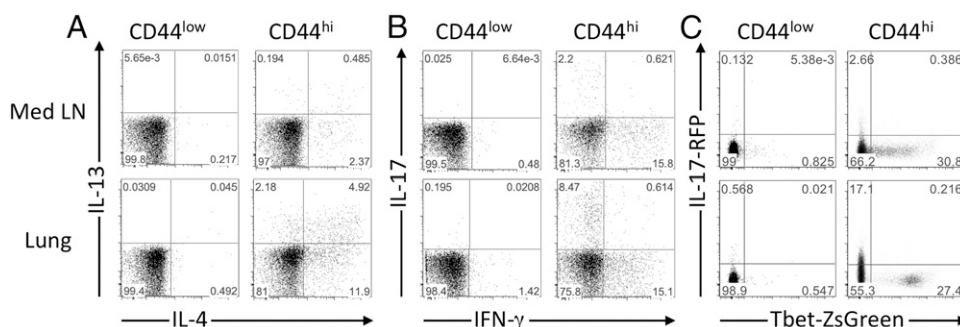


Fig. 2. Naïve phenotype lung CD4 T cells fail to produce cytokines or express key transcription factors. (A and B) CD4 T cells from mediastinal lymph nodes and lungs stimulated in vitro with phorbol 12-myristate 13-acetate and ionomycin and analyzed by intracellular staining for the production of IFN- γ , IL-17, IL-13, and IL-4. (C) Mice transgenic for Tbet-ZsGreen and IL-17-RFP were analyzed ex vivo for expression of ZsGreen and RFP without restimulation.

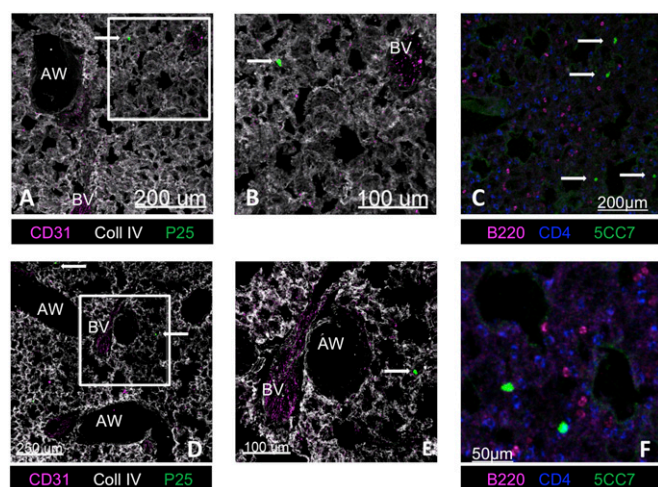


Fig. 3. Detection of naive transgenic CD4 T cells and endogenous CD4 in the lungs. Confocal microscopy images of frozen sections prepared from lungs of WT recipients of CD4⁺ cells from GFP⁺ Rag^{-/-} P25 mice (*A*, *B*, *D*, and *E*) or CFSE-labelled CD4⁺ from Rag^{-/-} 5C.C7 mice (*C* and *F*). Lungs were perfused and harvested at 18 h after transfer. In *A*, *B*, *D* and *E*, cells were stained with CD31 (purple) and collagen IV (white); transferred cells are GFP⁺. In *C* and *F*, staining was with anti-CD4 (blue), and B220 (purple); transferred naive T cells in green. AW, airway; BV, blood vessel.

FTY720 to explore the migratory patterns of lung CD4 T cells. FTY720 prevents lymphocytes from responding to the lymph/lymph node S1P concentration gradient, thereby sequestering them in the secondary lymphoid organs and leading to their loss from the blood and lymph (34–37). It also has been reported that FTY720 treatment depletes naïve T cells from the nonlymphoid organs and blocks the recruitment of antigen-sensitized CD4 T cells to the airway (10, 38).

We observed that in mice treated for 72 h with FTY720, lung CD4 T cells dropped to <3% of the level seen in untreated mice (Fig. 4A), whereas the numbers of lymph node naïve CD4 T cells were largely sustained (Fig. 4B). Flow cytometry analysis of lung

CD4 T cells remaining after 72 h of treatment showed that the majority of these cells had a “memory” CD62L^{low} CD44^{hi} phenotype, whereas the lymph nodes had a phenotype distribution resembling that of untreated mice (Fig. 4C). This finding implies that the lung draws its naïve cells from a compartment depleted by FTY720, presumably the blood, and that the naïve phenotype cells can leave the lung without following an S1P gradient.

To examine the kinetics of depletion, we transferred naïve 5C.C7 cells into B10.A mice, then started FTY720 treatment 24 h later. Half of the 5C.C7 cells were depleted from the blood within 4 h of the start of FTY720 treatment; comparable depletion of naïve CD4 T cells from the lung was observed at 18 h (Fig. 4A). This result implies that a period of ~14 h is required for naïve phenotype CD4 T cells to transit the lung. Along with giving insight into the dynamics of CD4 T cells in the lung, this result provides strong evidence that the naïve phenotype cells in the lung are not simply blood cells that have not been adequately flushed out.

Naïve CD4 T Cells Do Not Require L-Selectin or G Protein-Coupled Signaling to Enter the Lung.

lymph parenchyma from the blood is not clear. Treatment with anti-CD62L for 72 h had little effect on the frequency or phenotype of CD4 T cells in the lung, whereas it markedly diminished the number of CD4 T cells in peripheral lymph nodes, and the remaining cells were depleted of naïve phenotype cells. (Fig. 4*A, B*, and *D*) (39). Treatment with pertussis toxin (PT) blocks many G protein-mediated signals and thus the function of many chemokines (40). CFSE-labeled 5C.C7 cells and unlabeled 5C.C7 cells treated with PT were cotransferred into B10.A mice. Three days later, the frequency of PT-treated cells in the lung was similar to that in the blood and much greater than the frequency of untreated cells in blood or lung (Fig. 5*A*), implying that PT does not block the entry of naïve cells into the lung from the blood. When a mixture of WT and CCR7^{-/-} CD4 T cells was transferred, the blood-to-lung ratio was similar for both cell populations (Fig. 5*B*). Thus, entry into the lung appears to not depend on G protein-coupled responses.

Lung Naïve CD4 T Cells Appear to Supply a Substantial Proportion of Antigen-Responsive Naïve CD4 T Cells to Mediastinal Lymph Nodes.

Given that the entry of naïve CD4 T cells into the lung is not dependent on L-selectin, it might be anticipated that if these cells leave the lung through afferent lymphatics and migrate to

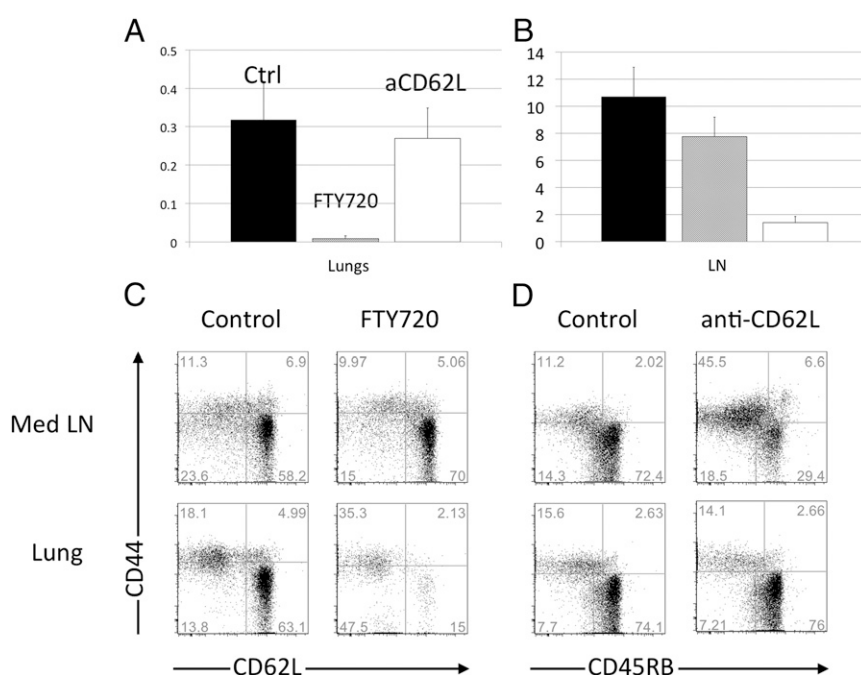


Fig. 4. (A and B) Effect of FTY720 and anti-CD62L on cells from lung (A) and mediastinal lymph nodes (B). Three groups are compared: normal mice (black), mice treated with FTY720 for 3 d (gray) and mice treated for 3 d with anti-CD62L (white). Cells were stained with anti-CD44 and anti-CD62L or anti-CD45RB. (C and D) Surface staining after 3 d of treatment with FTY720 (C) and anti-CD62L (D).

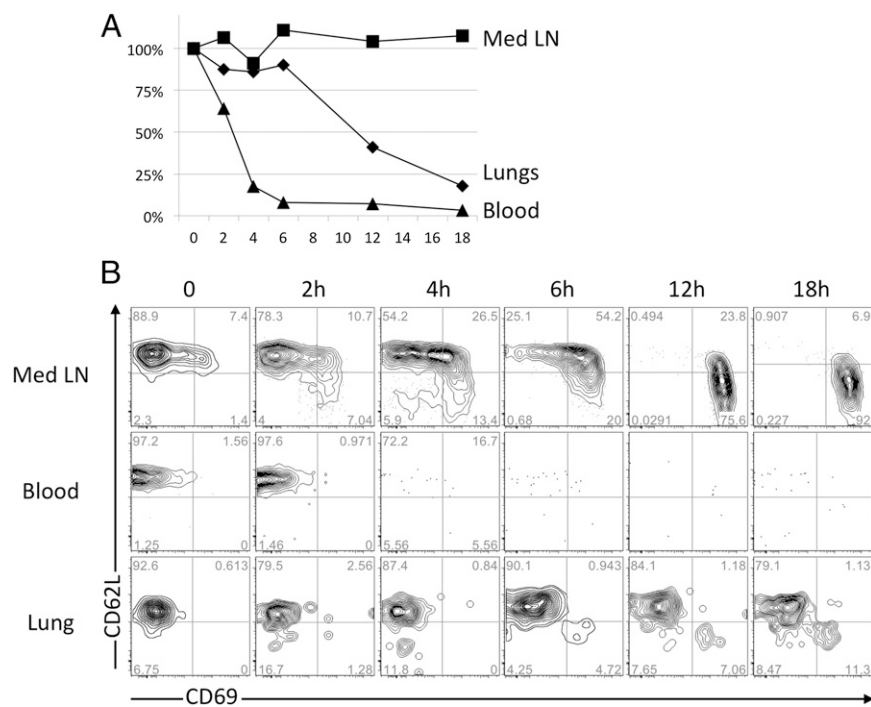


Fig. 6. Failure of naïve phenotype CD4 T cells to respond to antigen in the lung. B10.A (CD45.2) recipients of naïve CD45.1 5C.7 cells were immunized intratracheally with 100 μ g of PCC and 5 μ g of LPS. (A) Total numbers of naïve cells in blood, mediastinal lymph nodes, and lungs were determined at different time points after immunization; the results in the unimmunized controls were normalized to 100%. (B) Transferred 5C.7 T cells were analyzed for expression of CD62L and CD69 in the first 18 h after immunization.

failed to produce effector cytokines or to express the Th1 master regulator T-bet, based on the expression of a surrogate reporter, ZS-Green.

The frequency of naïve phenotype CD4 T cells in the lungs correlates with that in the blood, and the naïve phenotype cells in the lungs apparently are derived from the blood. However, it takes far longer to deplete cells from the lungs than from the blood after blockade of egress from the lymph nodes or antigen-mediated sequestration. Thus, whereas 60% of blood CD4 T cells are depleted within 4 h of FTY720 injection, a similar degree of depletion from the lungs takes 18 h, suggesting a mean lung transit time of ~14 h. A similar delay is noted when the depletion of TCR transgenic CD4 T cells from blood and lungs in response to antigen is measured.

Entry of naïve phenotype cells into the lungs is not inhibited by anti-CD62L, implying that these cells are not present in organized lymphoid tissue within the lungs. Microscopic analysis has shown that transferred naïve phenotype TCR transgenic cells are found in the lung parenchyma, not mainly within the vasculature (45). Entry into the lungs is not blocked by treatment of transferred cells with PT, although such treatment strongly blocks entry into lymph nodes (10, 43). Similarly, naïve phenotype CD4 T cells derived from CCR7-knockout donors, although unable to enter lymph nodes, normally gain access to the lungs. Thus, entry into the lungs appears to be largely independent of G protein-coupled receptors and does not require responsiveness to CCL19 or CCL21.

Despite their transit through the lungs, naïve CD4 T cells do not appear to respond to antigen in the lungs. TCR transgenic CD4 T cells remain in the lungs in diminishing numbers for ~18 h after intratracheal immunization, during which they retain their naïve phenotype. They do not up-regulate CD69, lose expression of Ly6C (a general marker of cell activation), up-regulate CD44, or down-regulate CD62L. In contrast, TCR transgenic CD4 T cells in the mediastinal lymph nodes show evidence of a response within 2 h of antigen challenge, and all of the TCR transgenic cells up-regulate CD44 and down-regulate CD62L by 18 h. It should be noted that lymphocyte activation in response to intratracheal immunization is limited mainly to cells in the lung-draining nodes. TCR transgenic cells are depleted from other peripheral lymph nodes, or, in mice treated with FTY720, nodes other than lung-draining nodes contain only naïve phenotype TCR transgenic cells, whereas the lung-draining nodes contain activated cells.

Our results strongly imply that the naïve phenotype cells that enter the lung, most likely from the blood, migrate to the lung-draining lymph nodes. Anatomic studies have identified the mediastinal lymph nodes as important lung-draining nodes (46). We have not directly established the migration of naïve phenotype lung CD4 T cells to lung-draining lymph nodes by visual inspection or by marking experiments; however, we have shown that naïve phenotype cells can enter the lung-draining lymph nodes through a CD62L-independent pathway under steady-state conditions. Treatment of mice with anti-CD62L decreased the frequency of naïve phenotype TCR transgenic cells in the popliteal lymph nodes by 90% within 6 h, compared with a 45% decrease in naïve phenotype CD4 T cells in the mediastinal lymph nodes, suggesting that approximately half of the naïve cells that gain entry to the mediastinal lymph nodes do so through a CD62L-independent pathway, presumably from the lungs. Although PT-treated and CCR7^{-/-} TCR transgenic cells enter the lungs normally, they fail to enter the mediastinal lymph nodes, implying that even the CD62L-independent component of naïve cell entry into these nodes is dependent on CCL19 or CCL21. TCR transgenic cells that do enter the mediastinal lymph nodes in mice treated with anti-CD62L are found within the CD4 T-cell region of the nodes and are capable of responding to antigen; that is, they demonstrate rapid induction of activation markers and by day 5 have adopted a full memory/memory effector phenotype.

What then is the value to the immune system of a blood-to-lung-to-lymph node circuit? It does not appear that the naïve phenotype cells initiate their immune response in the lung. No evidence of activation has been detected, and immunization through the lung was not found to induce a more prompt immune response than immunization by other routes. One attractive possibility is that this pathway may deliver more naïve CD4 T cells to the lung-draining nodes than normally would be brought to these nodes through the conventional blood/high endothelial venules route. In the early period after immunization, entry into lymph nodes remains CD62L-dependent, although later local inflammatory responses allow entry through other pathways, and the responding lymph nodes develop an increased blood supply. Assuming an 8- to 12-h transit time through lymph nodes, a typical naïve lymphocyte could make two or three circuits through the blood each day. If the rate of entry into the node is roughly proportional to the blood supply to

the node, and if in turn the relative blood supply to resting lymph nodes is correlated with the weight of the nodes, then it can be argued that a lymph node that constitutes ~5% of the weight of the total lymph node pool can sample only 10–15% of the naïve CD4 T cells on the first day after immunization. If that were the case, then only a small proportion of antigen-specific naïve cells would be capable of being activated during this period, and thus the immune response would be delayed. To the extent that a greater proportion of the lymphocyte pool could be sampled by the node in which a primary response was occurring, a more robust response might occur. Thus, the blood-to-lymph-to-mediastinal lymph node circuit may provide a second pathway to make naïve CD4 T cells available to the critical lymph nodes in which immune responses to lung challenges occur.

Materials and Methods

Mice. Animal care, handling, and experiments were performed in accordance with the guidelines of the National Institutes of Health's Animal Care and Use Committee. B10.A and 5C.7 TCR transgenic mice were purchased from Taconic. C57BL/6, B6.SJL CD45.1, CCR7^{-/-}, RAG-2pGFP, P25 TCR transgenic

mice were purchased from Jackson Laboratory. Further details are provided in *SI Materials and Methods*.

Adoptive Cell Transfer and Immunization. A total of 2×10^6 CD4 T cells were adoptively transferred in each recipient. Mice were immunized intratracheally with 100 μ g of PCC protein (Sigma-Aldrich) and 25 μ g of LPS (InVivoGen). For confocal microscopy studies, mice received CFSE-labeled MACS-purified CD4 T cells. In some experiments, mice were treated with 25 μ g of FTY720 (Calbiochem) or 250 μ g of anti-CD62L (MEL-14; BD Biosciences) injected i.p. Lymph node T cells were suspended in RPMI culture medium and incubated with 100 ng/mL PT per 3×10^7 cells for 2 h at 37 °C before being adoptively transferred (40). Further details are provided in *SI Materials and Methods*.

Flow Cytometry and Histology. Flow cytometry and histology analysis were performed following standard protocols for detection of intracellular antigens, as detailed in *SI Materials and Methods*.

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